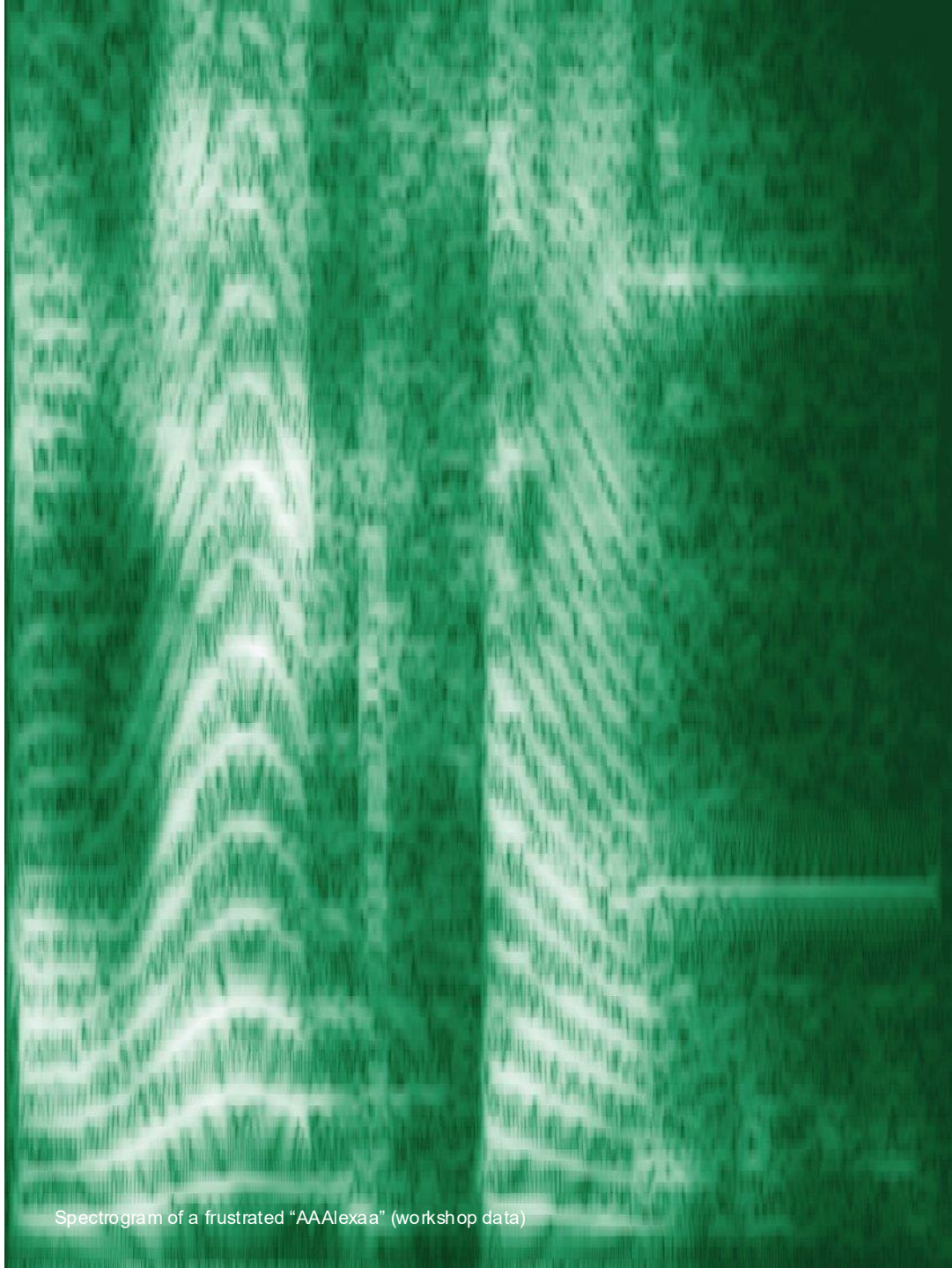
University of St.Gallen
Institute of Behavioral Science
and Technology

WORKSHOP · UNIVERSITY OF MELBOURNE

Speech Analytics

A hands-on introduction to speech analytics for behavioral research

Prof. Christian Hildebrand
University of St.Gallen (HSG) · Visiting Professor, CUHK
Wednesday, 11 November 2026 · 13:00–15:30



Spectrogram of a frustrated “AAAlexaa” (workshop data)

Brief note about myself

WHO

Role

- Full Professor of Marketing Analytics & Vice Dean, HSG
- Executive Director, Institute of Behavioral Science & Technology
- Visiting Professor, CUHK

WHAT

Research

- Consumer behaviour in interactions with AI
- Voice, conversational agents, agentic marketplaces
- New behavioural methods

HOW

Approach

- Paper → Model → Demo → Repo
- voiceR on CRAN (Busquet & Hildebrand)
- Open code for every workshop

My one-liner: I try to make the unmeasurable measurable, and voice is one of the richest places to do that.

Today’s agenda · Wed, 11 Nov 2026 · 13:00–15:30

13:00	Welcome & why voice (incl. live Alexa demo)	15′
13:15	Theory & measurement: speech, features, validity	40′
13:55	Break · open the notebook and test your audio	10′
14:05	Audacity warm-up: see and hear the differences	10′
14:15	Hands-on lab in R: calm vs. angry (Exercises 1–3)	40′
14:55	From analytics to voice agents · consent, privacy, bias	20′
15:15	Idea sprint (Exercise 4) & wrap-up	15′



By 15:30 you will be able to …

1

Argue for voice

Explain why voice is a distinct behavioural data source and where it adds value over text and self-report.

2

Map features to constructs

Link pitch, loudness, spectral, voice-quality and timing features to psychological states, and name the main validity threats.

3

Run a minimal pipeline

Load audio, plot waveforms and spectrograms, trim, extract features and compare conditions in R.

4

Design with voice

Sketch how voice enters your own studies as IV, DV or mediator, including via voice agents.

PART 1

Why voice?

Voice as a behavioural data layer: the how, not just the what

Background: spectrogram of a frustrated “AAAlexaa” (workshop data)

Motivation





Paralinguistics reveal various information about our emotions, personality as well as physicality (i.e., body size)

Elizabeth Homes
„Deep (Fake) Voice“

From self-reports to voice analytics

Traditional self-reports

- Depend on willingness & ability to report
- Ordinal scales (e.g., Likert) lack objectivity
- Retrospective and effortful

Voice analytics

- Capitalizes on physiological change during emotion
- Continuous, objective measurement
- Captures signal even without semantic content



Overview of what voice can tell us

MOMENTARY

States

- Emotion and arousal
- Stress and cognitive load
- Spontaneous reactions before reflection

STABLE

Traits

- Personality and confidence
- Dominance and competence
- Physicality (e.g., body size)

PERCEIVED

Attributions

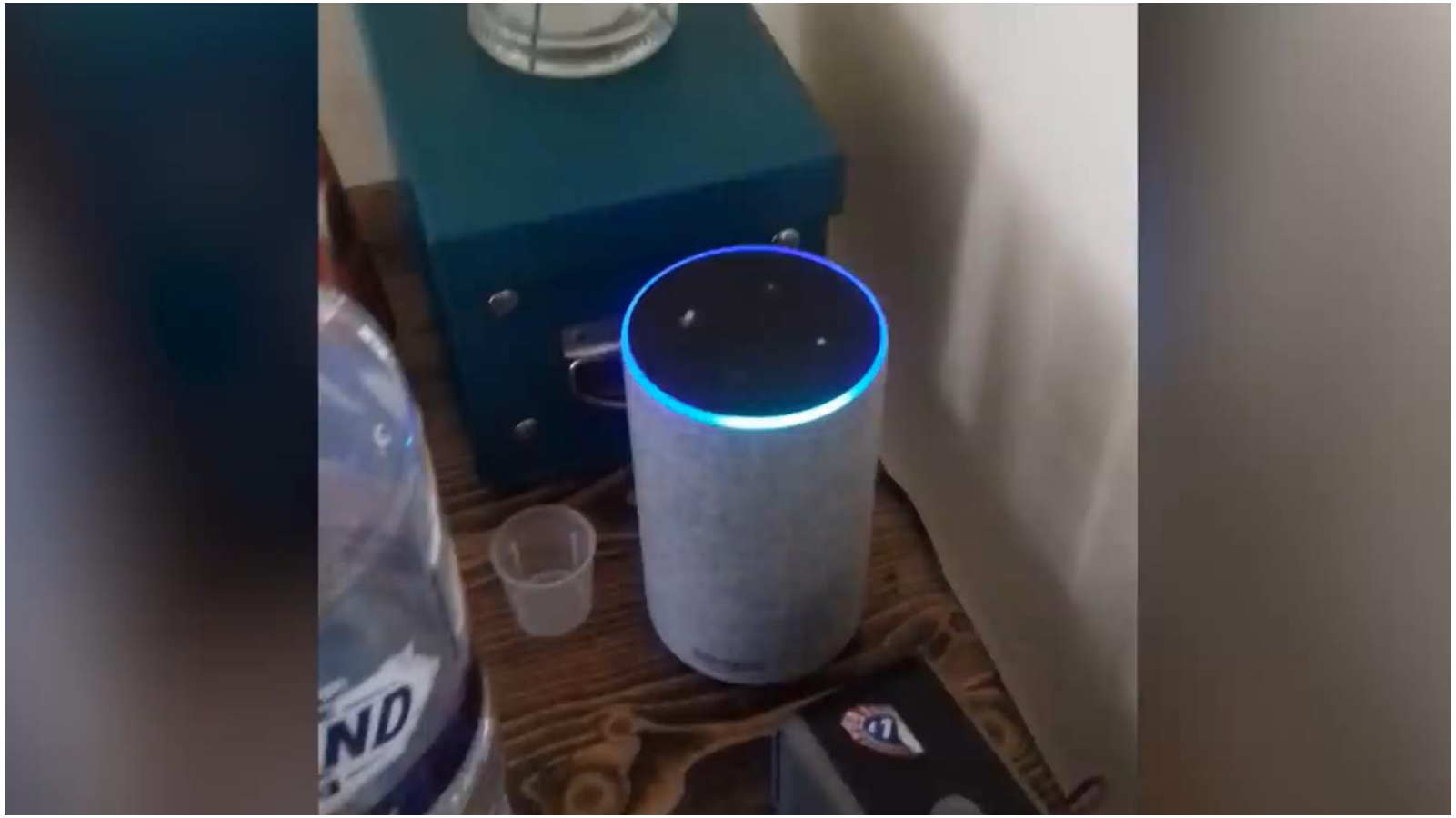
- Trust and warmth
- Persuasiveness
- Likeability of people and machines

Why now: smart speakers, voice assistants and LLM voice agents make voice data cheap to collect at scale, yet most behavioural researchers lack the tools to use it.

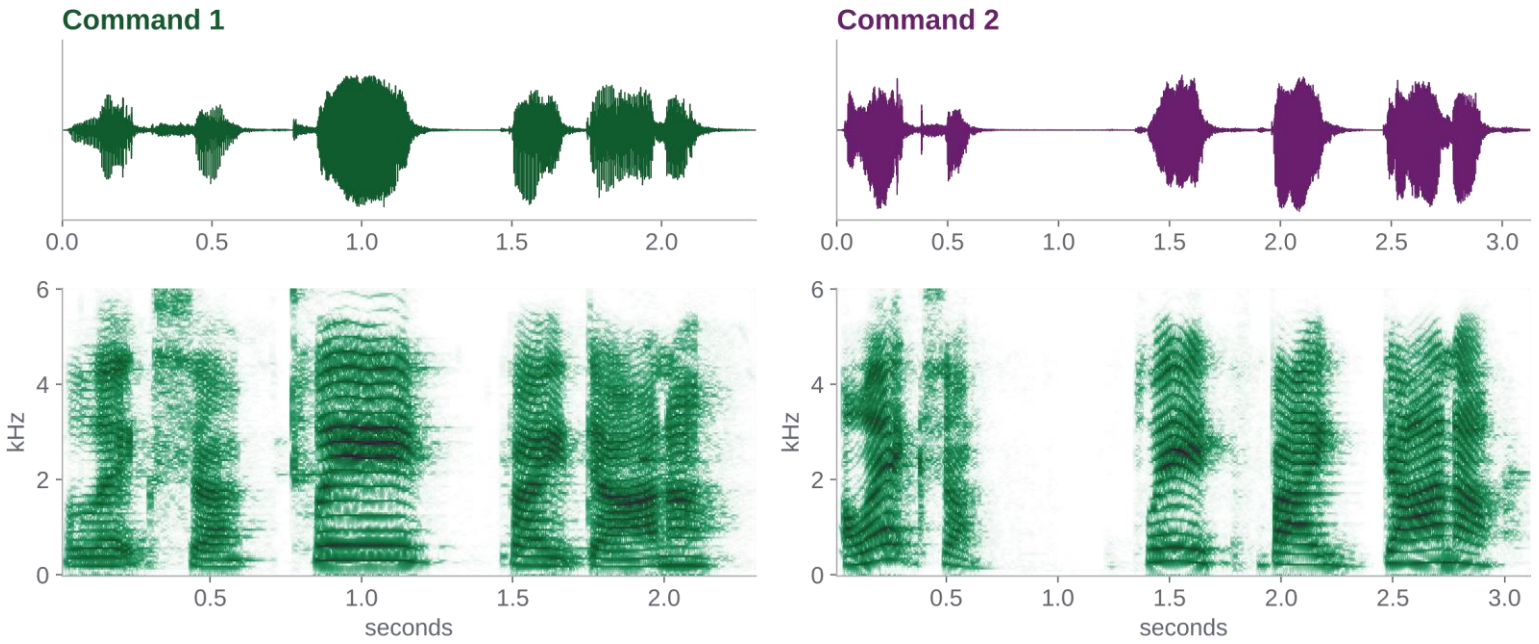
Live demo · sensing affective states from vocal cues

- How would you describe the **type of emotions** the user is experiencing?

- What exactly in the speaker‘s **voice marks** those emotions?



Live demo · same words, different meaning



EXTRACTED FEATURES

	Cmd 1	Cmd 2
Duration (s)	2.31	3.13
Vocal breaks	45%	52%
Loudness (sone)	16.2	18.1
Mean F0 (Hz)	222	271
Mean HNR	6.19	5.51

▶ Command 1
▶ Command 2

Source: Hildebrand, Efthymiou, Busquet et al. (2020), Journal of Business Research. Figure: spectrograms of the original workshop files.

Three ways to think about voice

INDEPENDENT VARIABLE

Voice as IV

Manipulate a vocal cue and measure an outcome.

E.g., does the vocal-tract length of an AI voice change product congruency and ad performance? (Efthymiou et al. 2023)

DEPENDENT VARIABLE

Voice as DV

Manipulate a condition and measure the voice.

E.g., does a failing voice assistant raise users' pitch and loudness? (Hildebrand et al. 2020)

MEDIATOR / MODERATOR

Voice as mechanism

Test whether a vocal feature carries an effect.

E.g., does vocal arousal mediate the effect of a service failure on complaint intentions?

Efthymiou, Hildebrand et al. (2023), The power of AI-generated voices; Hildebrand et al. (2020), JBR.

PART 2

Theory & measurement

From air pressure to psychological constructs, and where validity breaks

Background: spectrogram of a frustrated “AAAlexaa” (workshop data)

Theory foundations

Linguistics

A rigorous vocabulary for the signal

Phonetics

Source–filter model; feature definitions

Sociolinguistics

Variation is socially structured

Sociophonetics

Features carry social meaning & bias

Example

“REcord” (noun) vs. “reCORD” (verb): stress alone changes the meaning.

Example

Vocal folds set pitch (F0); the vocal tract filters it into formants that mark vowels.

Example

New York store clerks dropped more “r” in casual than careful speech (Labov, 1966).

Example

One speaker, two accents: listeners rate competence differently (matched-guise, Lambert 1960).

The speech chain: production → transmission → perception

Brain	→	Lungs & vocal folds	→	Vocal tract	→	Air, room & device	→	Listener or model
Intention, content and the motor plan for speech		The source: airflow and vibration set F0 (pitch)		The filter: tongue, lips and jaw shape formants		Transmission: distance, noise, mic and compression		Perception and inference about the speaker

Implication: a difference in your features can come from the speaker's state, their anatomy or your recording set-up. Your design has to tell them apart.

Four acoustic feature families

TIME

FEATURES

Duration, pauses, speech rate, % voiced

OFTEN SIGNALS

Hesitation, fluency, cognitive load

AMPLITUDE

FEATURES

Intensity, loudness, RMS energy, shimmer

OFTEN SIGNALS

Arousal, effort, dominance

FREQUENCY

FEATURES

F0 mean, range and SD, jitter, formants

OFTEN SIGNALS

Arousal, stress, emotion, size

SPECTRAL

FEATURES

HNR, spectral slope, centroid, entropy

OFTEN SIGNALS

Tension, roughness, harshness

Source: Hildebrand et al. (2020), Voice analytics in business research, Journal of Business Research 121, 364–374.

Each emotion has an acoustic signature

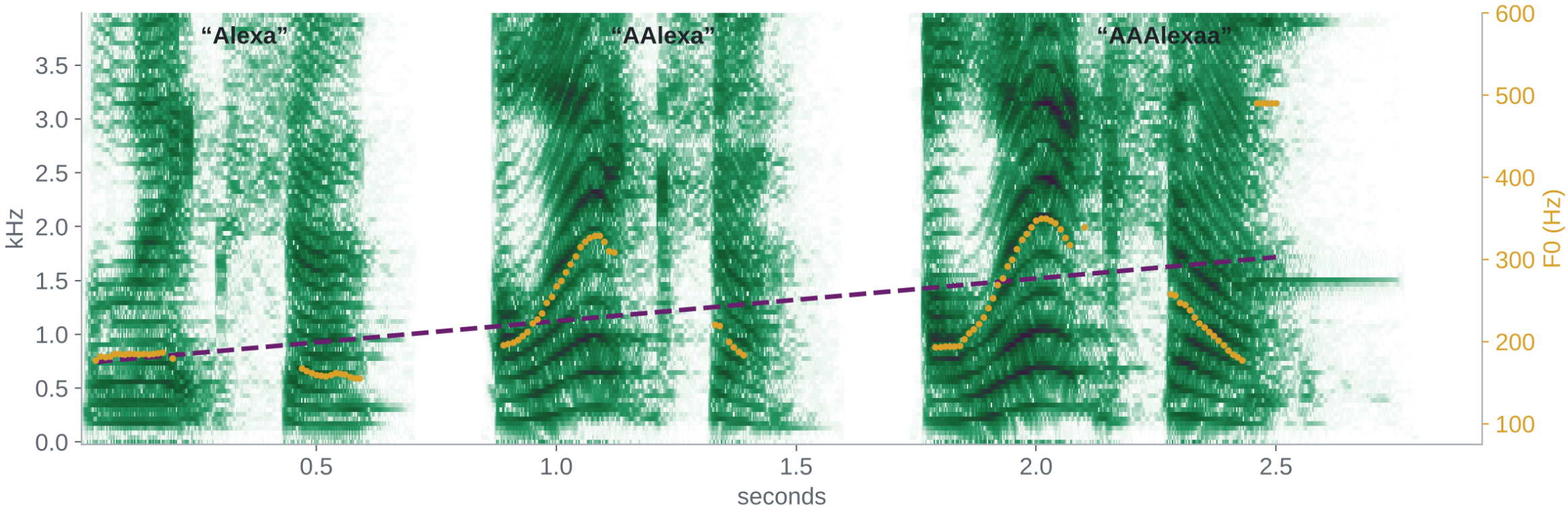
	Mean F0	F0 range	Intensity	Speech rate
Stress	High	–	High	High
Anger	High	–	High	–
Fear	High	High	–	–
Sadness	Low	Low	Low	Low
Joy	High	High	High	–
Boredom	–	Low	–	Low

READ IT LIKE THIS

- High-arousal states share a profile: high F0, high intensity
- Sadness and boredom sit at the low end
- Valence (anger vs. joy) is the hard problem

Stylized after Scherer et al. (2003) and Hildebrand et al. (2020).

Frustration you can see: recap of the three Alexa wake words



Gold: F0 track **Dashed:** linear F0 trend across repetitions

▶ 1 ▶ 2 ▶ 3

Mapping theory to measurement



Signal	Feature	Construct	Model	Inference
Pressure wave sampled over time	Numerical summary of the signal	Theoretical variable inferred from features	Statistical or ML link to outcomes	The claim you make, and for whom it holds

RULE OF THUMB

Never jump from a waveform to a psychological claim. Name the construct first, predict which feature families should move and in which direction, then check that they do.

Brief terminology review

1 Signal
Pressure wave sampled over time

2 Feature
Numerical summary of signal behavior

3 Construct
Theoretical variable inferred from features

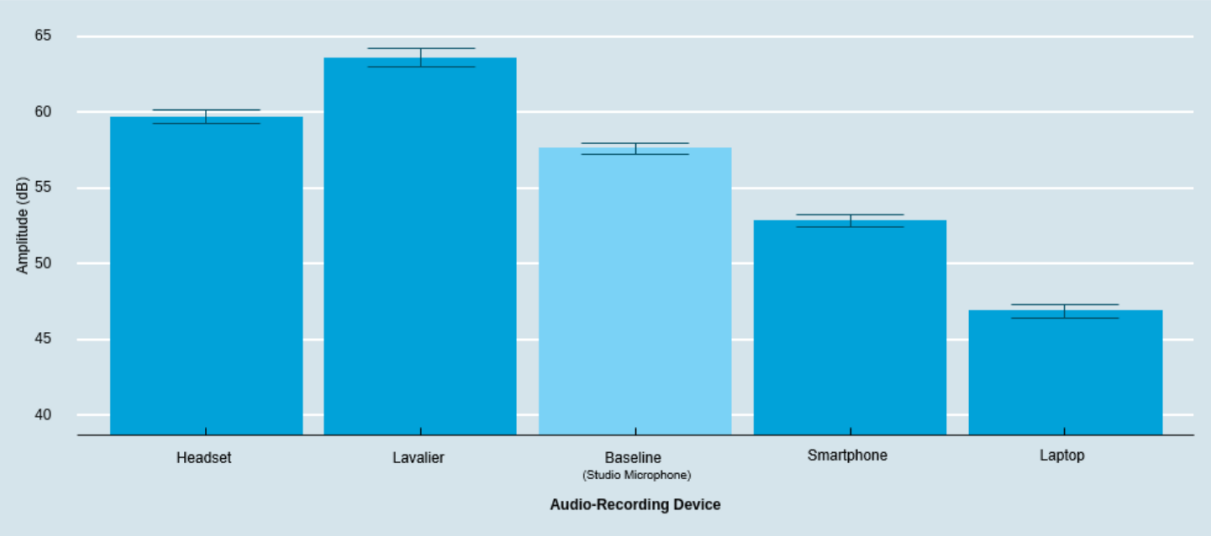
4 Model
Statistical or ML link to outcomes

5 Agent
Interactive system that collects or acts

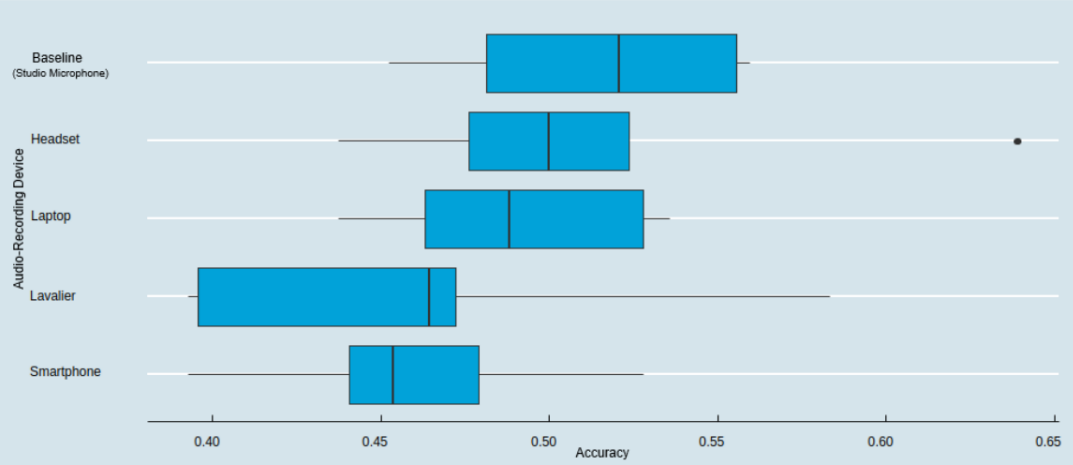
Rule of thumb

- Never jump straight from a waveform to a psychological claim.
- Write down the theory-to-measurement bridge before extracting features.
- Keep raw audio, derived features, transcripts, prompts, and logs linked by stable IDs.

Same voices, different devices, different data



Recorded amplitude (dB) by device

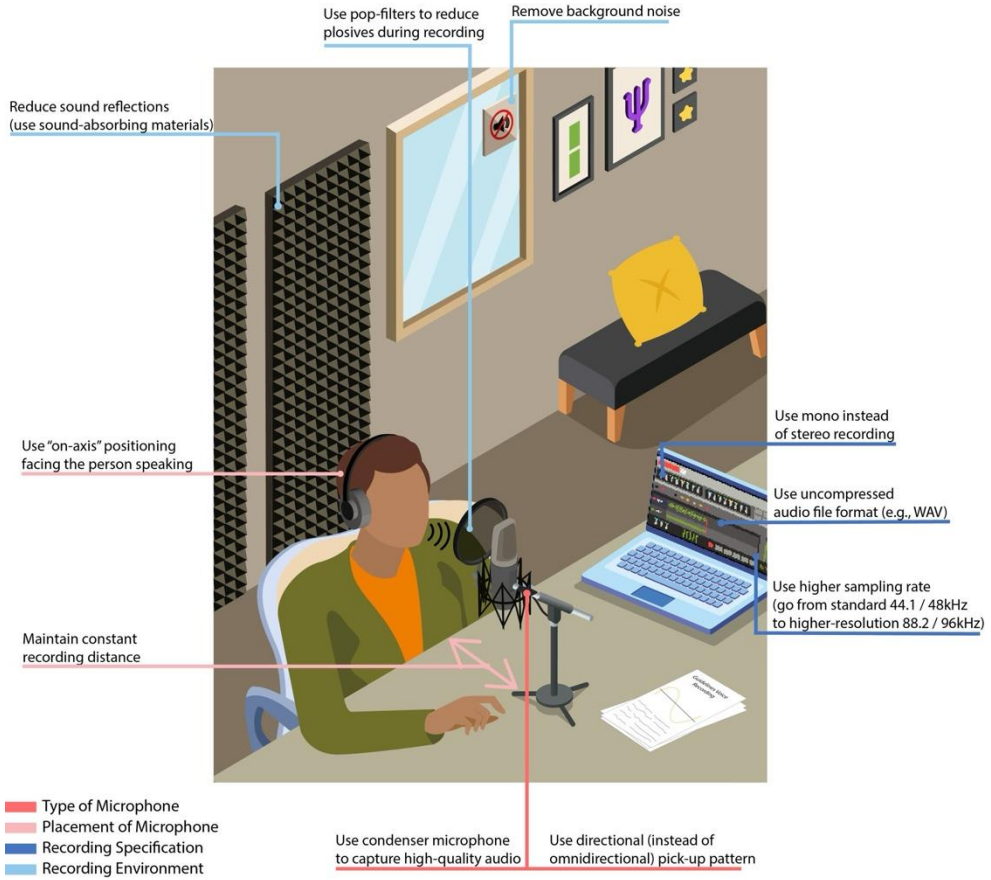


Emotion-prediction accuracy by device

- Standardise device, gain and distance within a study; never mix phones, laptops and headsets freely
- Record uncompressed WAV (≥ 44.1 kHz, mono) and store device as metadata

Busquet, Efthymiou & Hildebrand (2023), Voice analytics in the wild: Validity and predictive accuracy of common audio-recording devices, Behavior Research Methods.

Recording guidelines in one picture



DO

- Same device, gain, distance
- Uncompressed WAV, mono
- Quiet room, capture room tone
- Pilot and listen to every file

DON'T

- Mix devices across conditions
- Analyse lossy MP3
- Let the gain clip
- Denoise aggressively before extraction

Figure: recording guidelines from Busquet, Efthymiou & Hildebrand (2023), Behavior Research Methods.

Construct validity threats to think about

Device & room

Microphone, distance and compression shift amplitude and spectrum.

Speaker

Sex, age and accent change the baseline; vocal variation is socially structured.

Acted vs. spontaneous

Posed emotion is louder and clearer than the real thing.

Content & context

Words, language and task change the voice: hold them constant or model them.

Many features, few people

100+ features and 24 speakers invite overfitting: split by speaker.

Ask of every result

Would it survive a new microphone, new speakers and spontaneous speech?

PART 3

Hands-on lab

Listen → look → trim → extract → compare → predict

Background: spectrogram of a frustrated “AAAlexaa” (workshop data)

Audacity warm-up · see and hear the differences (10′)

1

Open & compare

Drag alexa_cmd1.wav and alexa_cmd2.wav into one project. Compare length, pauses, peaks.

2

Spectrogram view

Track menu → Spectrogram. Watch harmonics rise in the frustrated command.

3

Plot a spectrum

Select one vowel → Analyze → Plot Spectrum. Find F0 and the formants.

4

Edit, then re-check

Truncate Silence; change pitch by ±2 semitones and listen to what changes.

5

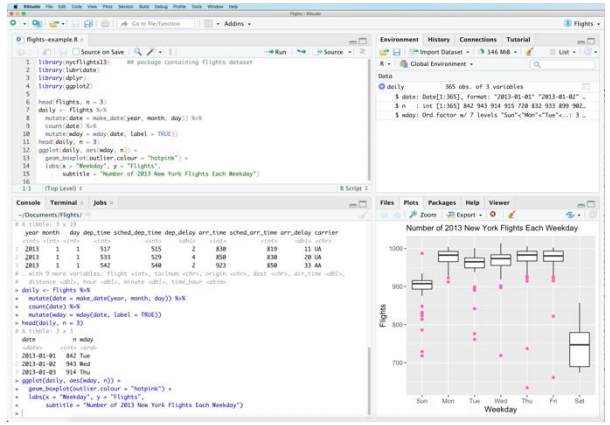
Hear the microphone

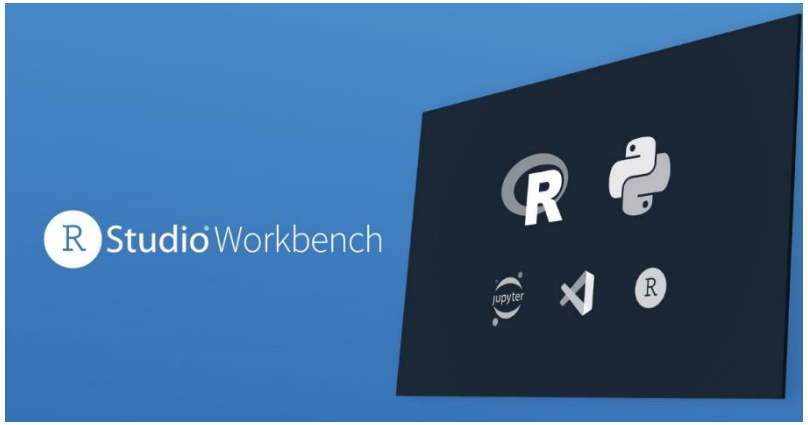
Load data/2_micTests: same speaker, 7 devices. Which roll off the highs?

Workshop site: childebrand.github.io/Melbourne-Voice-2026 **Free download:** audacityteam.org · menu names follow Audacity 3.x

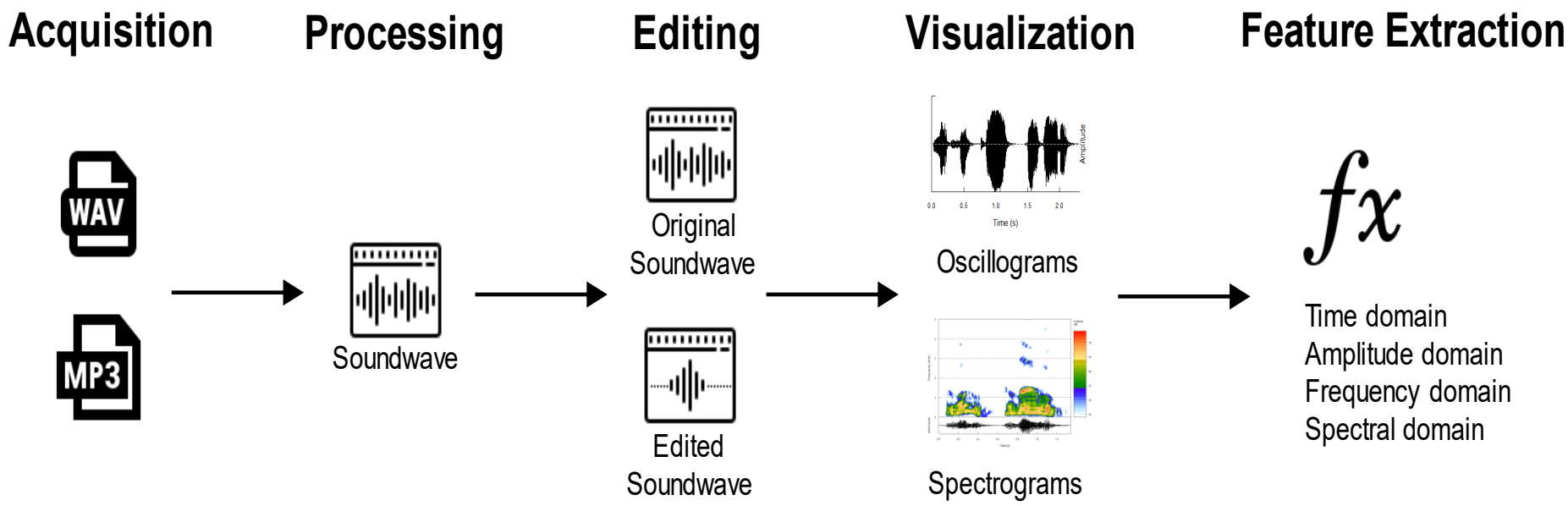
Why R for voice analytics?

- Automatic parallel or serial processing of many voice files
- A strong statistical-computing environment with an active community
- Mature audio packages: seewave, tuneR, soundgen, phonTools, bioacoustics
- voiceR package: one-line batch feature extraction + automated reports
- Seamless bridge to Python (Whisper, embeddings) through reticulate



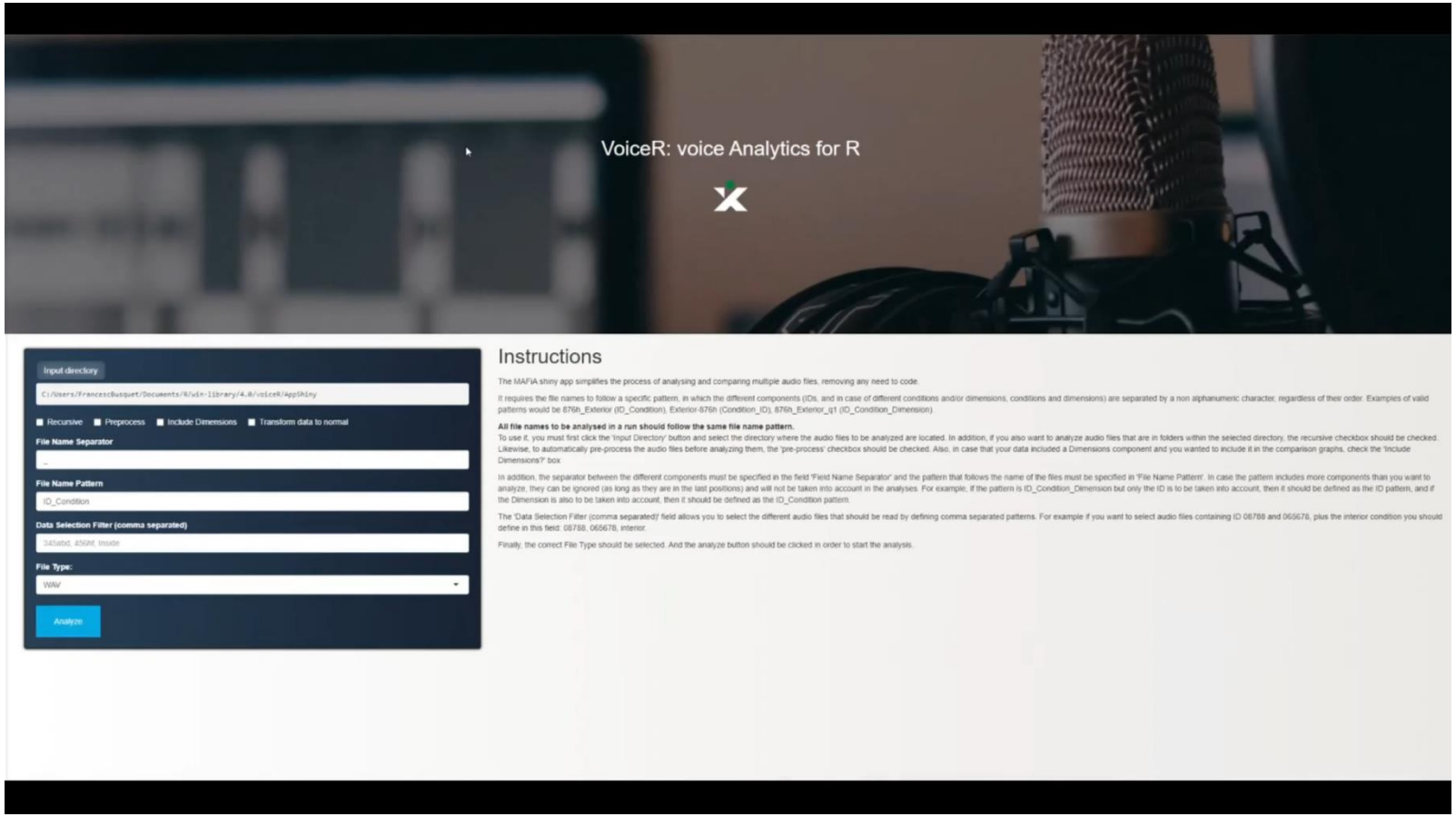


Voice analytics production pipeline



Source: Hildebrand et.al., 2020: Voice Analytics in Business Research.

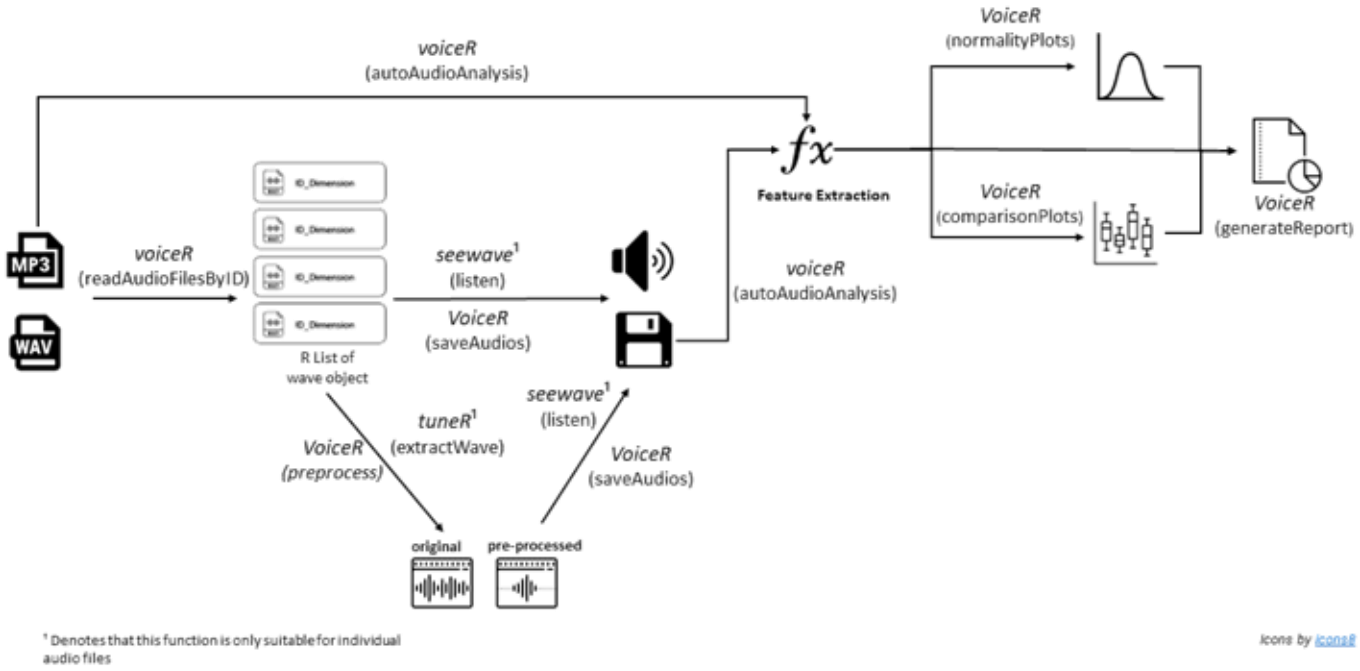
VoiceR package



voiceR

Package on Cran
(Busquet & Hildebrand 2023)

VoiceR package



voiceR
Package on Cran
(Busquet & Hildebrand 2023)

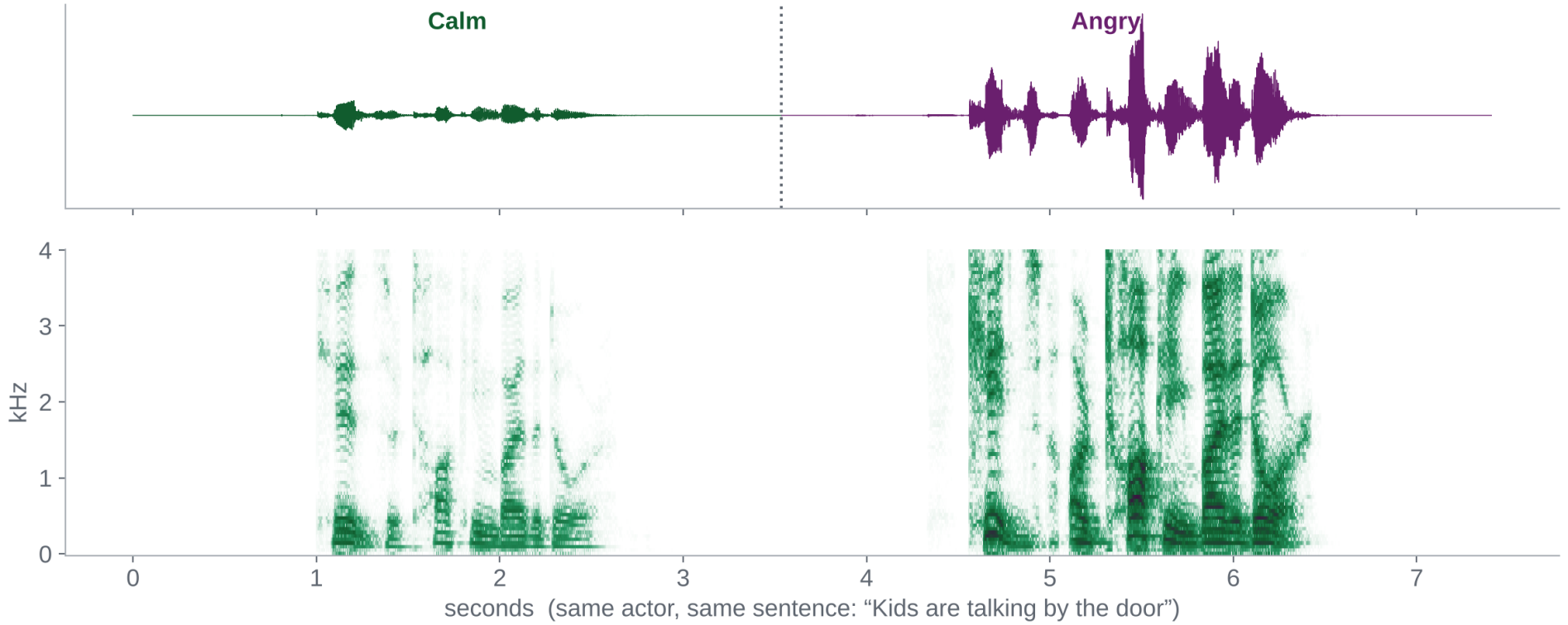
Lab roadmap · lab/voice_lab.qmd

0 · Setup	→	1 · Listen & look	→	2 · Trim	→	3 · Extract	→	4 · Compare	→	5 · Predict
Load packages, check audio (3′)		Oscillogram, spectrogram, bind() (Ex. 1)		timer() onsets, threshold check, extractWave()		voiceR::autoExtract() on 32 clips (Ex. 2)		Feature families by emotion		Speaker-independent model on 192 clips (Ex. 3)

THE CASE

RAVDESS (Livingstone & Russo 2018): 24 actors say “Kids are talking by the door” calm and angry, at two intensities. Same words, same actors, so the voice is the only thing that changes.

What you will see: calm vs. angry on one shared scale



▶ Calm ▶ Angry actor 01, normal intensity

Exercise 1 · Listen & Visualize

8 minutes

Pairs

Your task

1. Pick an actor (01–04). Plot calm vs. angry oscillograms and spectrograms side by side.
2. Now bind() the two clips and plot them on one shared scale. What appears that was hidden?
3. Run timer() with threshold 5 and 10. Which onset matches what you hear?

Deliverable

One sentence naming the cue that gives anger away first, and the onset threshold you would use.

Exercise 2 · Feature Extraction

12 minutes

Individual → pairs

Your task

1. Run voiceR::autoExtract() on the 32 trimmed clips (fileNamePattern given in the notebook).
2. Plot pitch_mean, pitch_sd, loudness_mean and HNR_mean by emotion.
3. Compare with the emotion-signature table: which predictions hold, which surprise you?

Deliverable

Which feature family separates calm from angry best, and one feature that behaved unexpectedly.

Exercise 3 · Prediction

10 minutes

Pairs

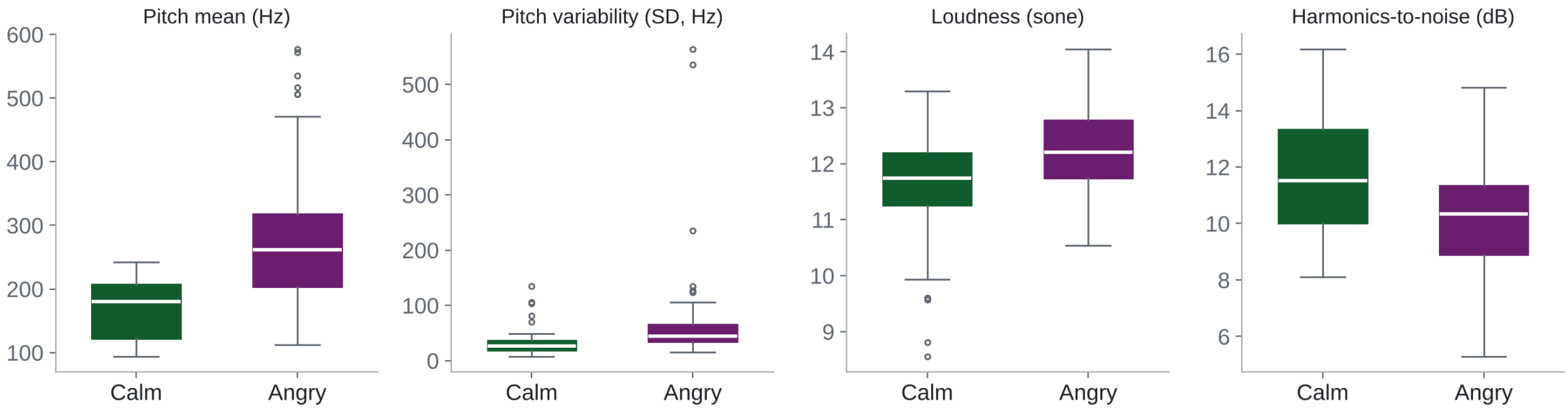
Your task

1. Fit a logistic or lasso model on all 192 clips, holding out whole actors.
2. Report test AUC. Then check female vs. male and normal vs. strong intensity.
3. Name the first thing that would break if you used this model on customer calls.

Deliverable

Your AUC, the weakest subgroup, and one validity threat for field use.

Debrief (all 192 audio clips)



- Anger raises pitch and pitch variability and lowers harmonics-to-noise: the arousal account holds
- Near-perfect accuracy here says more about acted speech than about anger detection in the wild

Features: voiceR::autoExtract(extended = TRUE) on silence-trimmed RAVDESS clips (Livingstone & Russo 2018).

PART 4

Extension: From analytics to voice agents

What is said, how it is said, and who gets to collect it

Background: spectrogram of a frustrated “AAAlexaa” (workshop data)

Speech-to-text adds the what to the how

Speech-to-text (ASR): end-to-end models such as OpenAI Whisper turn recordings into transcripts with timestamps, in R via the openai package.

Lexical

Word counts, type-token ratio, LIWC-style categories

Syntactic

Complexity, parts of speech, readability

Semantic

Sentiment, topics, emotion lexicons

Embeddings

Dense vectors for meaning; also audio embeddings for how it sounds

Jurafsky & Martin, Speech and Language Processing (3rd ed.); Berger et al. (2020), Uniting the tribes, Journal of Marketing.

Anatomy of a voice agent for research



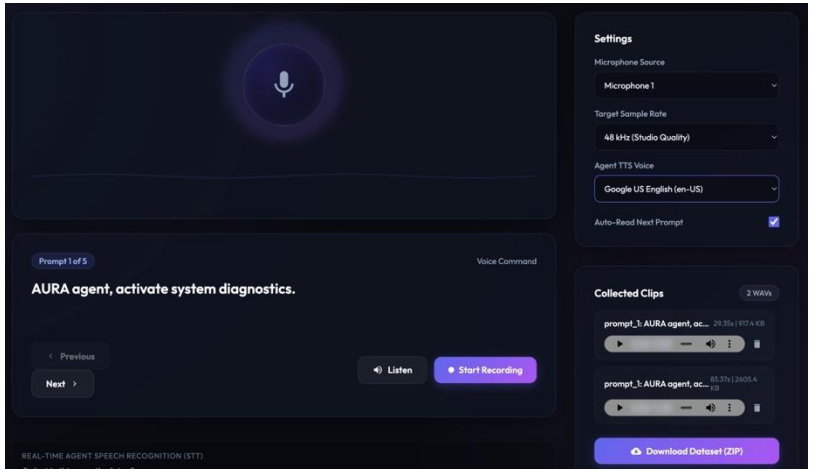
Use it to:

Run structured voice interviews and capture audio + transcript

Manipulate the agent's voice (pitch, rate, persona) as an IV

Log raw audio, transcript, latency, model, prompt and tool calls per turn

Fix success metrics and failure checks before you deploy



A simple agent prototyped with an AI coding tool (GSERM demo)

Consent, privacy and bias

PRIVACY

Voice is biometric

- Treat recordings as identifiable personal data
- Consent names capture, storage and analysis
- Store audio apart from identities

MINIMISATION

Sensitive content

- Speech can reveal health and other sensitive states
- Record only what the question needs
- Restrict and log access to raw audio

VALIDITY

Bias & fairness

- ASR and acoustic models can underperform for some accents, ages and genders
- Audit error rates by subgroup
- Report disaggregated performance

Exercise 4 · Idea sprint

15 minutes

Pairs → room

Your task

1. Pick one project of yours. Where could voice be the IV, the DV or the mediator?
2. Name the construct, the feature family that should move, and in which direction.
3. Name your data source (lab, online, field, agent) and the biggest validity threat.

Deliverable

A one-sentence pitch: “We will measure ___ in voice to show ___.”

Two or three pairs share.

Wrap-up & resources

Today's take-aways

- Voice is a behavioural data layer: the how, not just the what
- Map feature families to constructs with theory first
- Device, speaker and acted speech are the big validity threats
- The pipeline is simple: read, look, trim, extract, model, check subgroups

Keep going

Course website: childebrand.github.io/Melbourne-Voice-2026

voiceR: autoExtract(), autoReport(); install via setup script (CRAN archive)

Read: Hildebrand et al. (2020) JBR; Busquet et al. (2023) BRM; Hildebrand, Cascio Rizzo & Berger (2026) COPSYC

Full RAVDESS: scripts/get_full_ravdess.R

Thank you

Go listen to your data.

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